

1. Project overview

This project implements core technical-analysis calculations in R using custom functions and base R operations. The assignment specifically requires the indicators to be implemented without relying on ready made technical analysis library functions. Each indicator is stored in its own R script so that it can be tested independently.

Repository contents

File	Implementation
sma.R	Simple Moving Average
ema.R	Exponential Moving Average
macd.R	Moving Average Convergence Divergence
stdev.R	Standard Deviation
linreg.R	Linear Regression
rsi.R	Relative strength Index
stoch_rsi.R	Stochastic RSI
crossover.R	Crossover Signals
crossunder.R	Cross under Signals

How to run the project in R studio

1. Open R studio and open or download the BDA400-Assignment-5 repository folder.
2. Open each R script and click source to run the complete file. Each script includes sample data and a print statement for verification.
3. Run sma.R before stoch_rsi.R, because Stochastic RSI uses the sma() function. Run rsi.R before stoch_rsi.R because it also uses rsi().
4. Run ema.R before macd.R, because macd() uses the ema() function.
5. No external technical-analysis packages are required for these implementations.

Recommended Source Order

```
Source("sma.R")
Source("ema.R")
Source("macd.R")
Source("stdev.R")
Source("linreg.R")
Source("rsi.R")
Source("Stoch-rsi.R")
Source("crossover")
Source("crossunder.R")
```

2. Indicator Implementation

1. Simple Moving Average (SMA)

R file = sma.R
Function = sma(data,period)

PURPOSE: Calculate an equal-weight moving average over a selected number of observations.

IMPLEMENTATION: The function checks that enough data is available, creates an output vector, moves through each window of the requested period, sums the values in the window and divides by the period.

OUTPUT: A numeric vector containing the SMA values for each complete window.

SAMPLE VERIFICATION: For data 10,12,15,20,18,22,25,24,21 the result begins 12.33333, 15.66667, 17.66667

2. Exponential Moving Average (EMA)

R file = ema.R
Function = ema(data,period)

PURPOSE: Calculates a moving average that gives more weight to recent observations.

IMPLEMENTATION: The multiplier is calculated as $2/(period+1)$. The first EMA is initialized with the first data value. Each later EMA is calculated from the current value, the previous EMA and the multiplier.

OUTPUT: A numeric vector with one EMA value for each input observation.

SAMPLE VERIFICATION: With period 3, the sample output starts 10.00000, 11.00000, 13.00000, 16.50000.

3. Moving Average Convergence Divergence (MACD)

R file = macd.R
Function = macd(data, short_period, long_period, signal_period)

PURPOSE: Measures the relationship between a short-term EMA and a long-term EMA and produces a signal line and histogram.

IMPLEMENTATION: The function calculates short and long EMAs, subtract them to obtain the MACD line, applies EMA again the signal line and subtract the signal line from the MACD lines to produce the histogram.

OUTPUT: A list containing MACD_line, Signal_line and Histogram.

DEPENDENCY: Requires ena() to be loaded first.

SAMPLE VERIFICATION: the RStudio test returned all three list components successfully.

4. Standard deviation

R file = stdv.R
Function = stdv(data)

PURPOSE: measure the spread or variability of a numeric series

IMPLEMENTATION: the implementation manually calculates the mean, differences from the mean, squared differences, population variance using N in the denominator, and finally the square root of the variance.

OUTPUT: A single numeric standard deviation value.

SAMPLE VERIFICATION: for the sample data, the calculated values are approximately 4.946629.

5. Linear regression

R file = linreg.R
Function = linreg(regression source, regression length, regression offset)

PURPOSE: calculates a simple least square regression line for a selected portion of a numeric series.

IMPLEMENTATION: the function validates the regression length and offset, selects the required data a subset, creates index values, calculates their means, computes the regression numerator and denominator, and then calculates slope, intercept and predicted values.

OUTPUT: a list containing slope, intercept and predicted values.

SAMPLE VERIFICATION: the sample test reduces slope 0.7428571, intercept 19.06667 and a vector of predicted values.

6. Relative strength index (RSI)

R file = rsi.R
Function = rsi(data,period)

Purpose: measures price momentum by comparing average gains and average losses

IMPLEMENTATION: the function calculates consecutive price differences, separates gains and losses, calculates initial average over the selected., and length applies Wilder style smoothening. RSI has calculated as $100 - 100 / (1 + RS)$

OUTPUT: a numeric vector containing NA values until enough history exists, followed by RSI values from 0 to 100

SAMPLE VERIFICATION: for the sample period of five, the calculated values include 50.52632, 68.92567, 71.83521, 76.04142 and 71.58684

7. Strategic RSI

R file = stoch_rsi.R
Function = stoch_rsi (data,period, K_period, D_period)

PURPOSE: Normalizes RSI values and produces % K and %D lines for momentum analysis.

IMPLEMENTATION: The function first calls RSI(), finds the minimum and maximum valid RSI values, normalize the RSI series, applies sma() to create the %K line, and applies sma() again to create the %D line.

OUTPUT: A list containing K_line and D_line.

DEPENDENCY: Requires rsi() and sma() to be loaded first

SAMPLE VERIFICATION: The sample test produces valid %K values and %D values after the required initial NA periods.

8. Crossover function

R file = crossover.R
Function = crossover (arr1, arr2)

PURPOSE: Detects when one series crosses above or below another series.

IMPLEMENTATION: The function first confirms both arrays have the same length. Beginning with the second observation, it compares the current and previous values of both arrays. It returns up when arr1 crosses above arr2, down when it crosses below, and none otherwise.

OUTPUT: A character vector containing up, down or none signals.

SAMPLE VERIFICATION: For the provided samples, up signal appears at the fourth position.

9. Cross under function

R file = crossunder.R
Function = crossunder (arr1, arr2)

PURPOSE: Detects a specific downward crossing event where arr1 moves from at above arr2 to below arr2.

IMPLEMENTATION: After validating equal array length, the function checks each point from the second observation onwards. If $arr1[i] < arr2[i]$ and $arr1[i-1] \geq arr2[i-1]$, the position is marked true otherwise it is marked false. The first position is initialized as none.

OUTPUT: A character vector containing none, true or false.

SAMPLE VERIFICATION: For the assignment sample, the output contains none in the first position and false for the remaining positions.